Automatic Street Light (ON/OFF)

Arduino Uno + Tinkercad Simulation Project

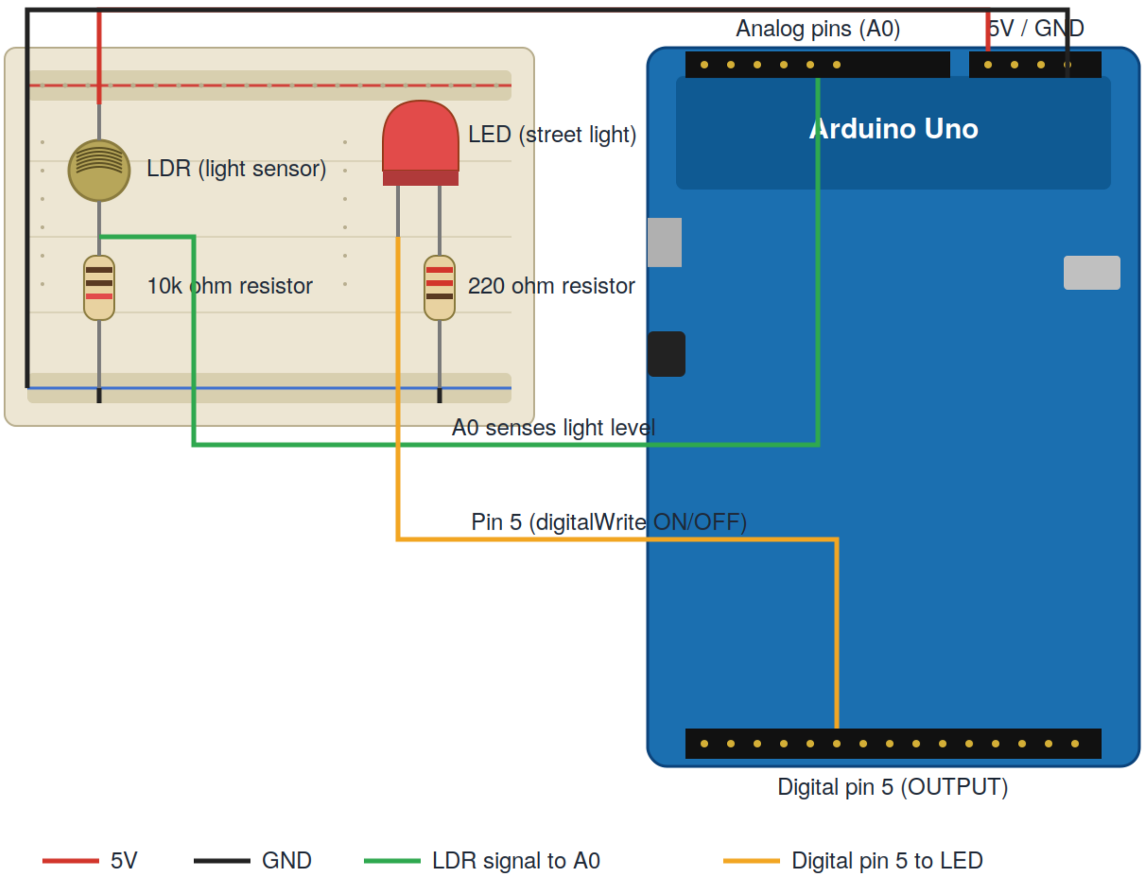
Project Overview

Yeh project ek LDR (photoresistor / light sensor) ka istemal karke andhera detect karta hai, aur andhera hone par ek LED (jo street light ko represent karti hai) ko automatically ON kar deta hai. Roshni hone par LED wapas OFF ho jati hai. Ismein LED sirf full ON ya full OFF hoti hai (koi dimming nahi) - decision Arduino code mein ek if/else condition se hota hai.

Components Used

#	Component	Quantity	Purpose
1	Arduino Uno R3	1	Main microcontroller board
2	Breadboard (small)	1	To hold and connect components
3	Photoresistor (LDR)	1	Senses ambient light level
4	Resistor - 10k ohm	1	Voltage divider with LDR (for A0 reading)
5	LED	1	Represents the street light
6	Resistor - 220 ohm	1	Current-limiting resistor for the LED
7	Jumper wires	as needed	Connections between components

Circuit Diagram (Tinkercad Style)



LDR voltage divider wired to A0 (green wire) and LED wired to digital pin 5 through a 220Ω resistor (orange wire). Red = 5V, Black = GND.

Step-by-Step Build Procedure

Step 1: Naya circuit banao

Tinkercad kholo, "Circuits" section mein jao, aur "Create new Circuit" par click karo.

Step 2: Components add karo

Right-side panel se ye 5 cheezein drag karke canvas par lao: Arduino Uno R3, Breadboard (small), Photoresistor (LDR), Resistor (do adad - 10k ohm aur 220 ohm), aur LED.

Step 3: LDR ka voltage divider banao

LDR ki ek leg breadboard ke ek row mein, dusri leg alag row mein lagao. LDR ki upar wali leg ko Arduino ke 5V pin se wire karo. LDR ki nichay wali leg ko 10k ohm resistor ke through GND se jodo. LDR aur resistor ke beech wale junction point ko Arduino ke A0 (analog) pin se wire karo.

Step 4: LED wiring karo

LED ka anode (lambi leg) Arduino ke pin 5 se wire karo. LED ka cathode (choti leg) 220 ohm resistor ke through GND se jodo.

Step 5: Sab GND ko common karo

Breadboard ki GND rail ko Arduino ke GND pin se connect karo, taake LDR aur LED dono ka ground same ho.

Step 6: Code likho

Arduino ko select karo, "Code" button dabao, "Blocks + Text" ki jagah Text mode select karo, aur neeche diya gaya pura sketch paste karo (dekhein agla section).

Step 7: Simulation chalao

"Start Simulation" button dabao. LDR par click karke uski light-level slider ghumao - jab roshni kam ho (dark), LED ON ho jayega; jab roshni zyada ho, LED OFF ho jayega.

Step 8: Threshold check aur adjust karo

Simulation ke waqt Serial Monitor kholo, LDR ki values dekho. Agar din (bright) mein values 700-900 aur raat (dark) mein 100-300 ke around aayen, to threshold = 500 theek rahega. Agar range different ho to code mein threshold value us hisaab se badal dena.

Complete Arduino Sketch

```
// C++ code
int LightSensor = 0;
int threshold = 500;   // is value se neeche = raat/andhera samjho

void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);
  pinMode(5, OUTPUT);
}

void loop()
{
  LightSensor = analogRead(A0);
  Serial.println(LightSensor);

  if (LightSensor < threshold) {
    // andhera hai (raat) -> light ON
    digitalWrite(5, HIGH);
  } else {
    // roshni hai (din) -> light OFF
    digitalWrite(5, LOW);
  }

  delay(10);
}
```

Wiring Summary

From	To	Wire Color (suggested)
Arduino 5V	LDR - leg 1	Red
LDR - leg 2	10k resistor - leg 1	Direct breadboard row
LDR - leg 2 / Resistor junction	Arduino A0	Green
10k resistor - leg 2	GND rail	Black

Arduino Pin 5	LED anode (long leg)	Orange
LED cathode (short leg)	220 ohm resistor - leg 1	Direct breadboard row
220 ohm resistor - leg 2	GND rail	Black
Breadboard GND rail	Arduino GND	Black

Kaise Kaam Karta Hai

LDR aur 10k resistor mil kar ek voltage divider banate hain, jiski beech wali value Arduino ke A0 pin par padhi jati hai (analogRead). Jab roshni kam hoti hai (andhera), LDR ki resistance badh jati hai aur A0 par value kam ho jati hi. Code is value ko threshold (500) se compare karta hai: agar value threshold se kam ho to Arduino pin 5 ko HIGH kar deta hai, jisse LED ON ho jati hai. Agar roshni zyada ho (value threshold se zyada) to pin 5 LOW ho jata hai aur LED OFF rehti hai. Yeh poora check har 10 milliseconds (delay(10)) mein dobara hota hai, isliye light automatically adjust hoti rehti hai.

Note: LED ko theek se on karne ke liye A0 aur pin 5 dono ka connection sahi hona chahiye - agar simulation mein LED ka behavior ulta lage (roshni mein ON, andhera mein OFF), to code mein **<** ko **>** se replace kar dena is line mein: if (LightSensor < threshold).

Tinkercad Circuit Screenshot